INES ALONSO

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Research interests: *stochastic dynamics of molecular and learning systems; kinetic models; verifiable ML for the physical sciences*

EDUCATION

Florida Polytechnic University — *Lakeland, FL, USA*
B.S. Computer Science, Minor in Applied Mathematics · GPA 3.95/4.0 — **Expected Dec 2026**

University of Barcelona — *Barcelona, Spain*
Undergraduate physics coursework (three semesters); transferred to Florida Polytechnic — **Transferred 2024**

Relevant coursework: Machine Learning, Natural Language Processing, Quantum Information & Computing, Numerical Methods, Computational Linear Algebra, Algorithm Design & Analysis, Multivariable Calculus.

Honors: SPIE@ICFO María Yzuel Fellowship; Florida Polytechnic Merit Scholarship; Esperanza Inc. Scholarship; President's List (3 terms); IBM Qiskit Global Summer School Quantum Excellence Badge.

RESEARCH EXPERIENCE

Institute of Photonic Sciences (ICFO) — *Barcelona, Spain*
Research Intern in Computational Biophysics, SPIE@ICFO María Yzuel Fellow — **May 2025 – Present**
Supervisors: Prof. Nicoletta Liguori (Photon Harvesting in Plants and Biomolecules) and Prof. Maciej Lewenstein (Quantum Optics Theory)

- Designed and ran the full kinetic analysis of **175 coarse-grained (Martini 2.2) LHCII trajectories, 4,500 µs of sampling** across three pH/PsbS conditions (MDTraj, deeptime, PyEMMA).
- Built an **out-of-core pipeline** streaming TICA covariance estimates over all **26,796 inter-residue distances** (~750,000 frames, ~80 GB) with no downsampling or feature pruning; trained a five-layer **VAMPnet** (PyTorch) to learn nonlinear slow coordinates.
- Built and stress-tested **Markov state models two independent ways** (clustering-based and physics-defined states) with Chapman–Kolmogorov tests, spectral-gap analysis, held-out cross-replica validation, and designed known-zero and false-positive controls.

Florida Polytechnic University, Physics Department — *Lakeland, FL*
Quantum Software Research Assistant · Advisor: Dr. Bradford A. Barker — **Feb 2024 – Present**

- First author and co-developer of **q-sc-SF-EOM**, a quantum self-consistent spin-flip equation-of-motion algorithm that obtains ground and excited states from one eigenvalue problem built on a high-spin reference (Qiskit, PySCF, ADAPT-VQE).
- Cut circuit cost at equal accuracy: the costliest (2e,2o) matrix element drops from **52 to 16 CNOTs**; for LiH, a spin-completed pool stays **within 0.1 eV of FCI at 3.3× fewer CNOTs** than conventional q-sc-EOM.
- Developed a **completeness analysis** of how much of the $M_S = 0$ determinant space each operator pool reaches (100% in (2e,2o), 40% for LiH (2e,5o), 25% for H_4), explaining when and why single flips yield spin-contaminated states across five molecules.

Agentic MSM Benchmark Engine (independent) · github.com/inesalonsoo/multiagent_for2Dphysics — **Jul 2026 – Present**

- Built a **three-agent LLM system** (Orchestrator / Optimizer / Validator) that proposes Markov state model analyses; results are accepted only by **deterministic checks against exact physics**, never by the LLM; ground truth formalized in **Lean 4** (7 statements, 2 proved, one with Axiomatic AI's ax-prover).
- Validated the MSM engine against the **exact relaxation rate** of a stochastic double well (computed from the discretized generator): two metastable states found in **42/42 replicas**; measured/exact rate **0.975–1.031 for β = 3–7**.
- **Audited my own pipeline:** the textbook Eyring–Kramers gate is **6–10% off** at these temperatures, so the loop had accepted biased analyses; replaced it with an exact oracle and reported overconfident Bayesian error bars (**2 of 42** contain the true rate).

PUBLICATIONS & PRESENTATIONS

- **I. Alonso**, W. J. Hall, S. Mandulapalli, D. A. Strubbe, B. A. Barker, "The q-sc-SF-EOM quantum algorithm for ground- and excited-states in molecules and solids," manuscript in preparation.
- Poster, q-sc-SF-EOM, IEEE Quantum Week (QCE26), Toronto, Canada, Sep 2026.
- Lightning Round podium talk and poster (selected, 20 of 80 posters), NanoFlorida 2026, Lakeland, FL, Oct 2026.
- Selected poster, STEM Research Competition (Undergraduate), SHPE National Convention, Indianapolis, IN, Oct 2026.

ADDITIONAL PROJECT

Hierarchical Reasoning Model with Attention Feedback — **Nov 2025**

- Designed the token-level module of a 60.6M-parameter Hierarchical Reasoning Model variant for multi-hop QA on HotpotQA, in which sentence- and token-level modules exchange information through bidirectional cross-attention (4-person team, NLP course, PyTorch).

LEADERSHIP

Sustainability President, Student Government Association, Florida Polytechnic — **May 2025 – Jul 2026**

- Founded the Sustainability Agency; led 8 ambassadors, built a 100 m² campus garden ($2,500 budget), ran 200+ attendee events.

Vice President of Internal Affairs, Society of Hispanic Professional Engineers (SHPE) — **Jan 2025 – Jul 2026**

- Ran 10 professional-development workshops and 8 heritage events; liaison to the Provost's Office and Career Services.

SKILLS

Programming: Python, C/C++, SQL, Lean 4, LaTeX, Git, Linux, Docker, REST APIs, Pydantic
ML & scientific: PyTorch, scikit-learn, deeptime, PyEMMA, MDTraj, NumPy/SciPy, Qiskit, PySCF, CUDA, MATLAB, LLM APIs
Languages: Spanish (native), Catalan (native), English (fluent)